Impossibility of a 3×3 Magic Square of Distinct Squares

Dependent Variables and the Dimension Restriction

Fix the center of the 3×3 magic square to be C^2 and write the remaining entries as $C^2 + r_i$:

$$\begin{array}{ccc} C^2 + r_1 & C^2 + r_2 & C^2 + r_3 \\ C^2 + r_4 & C^2 & C^2 + r_5. \\ C^2 + r_6 & C^2 + r_7 & C^2 + r_8 \end{array}$$

The magic condition requires that every row, column, and diagonal satisfy

$$r_a + r_b + r_c = 0.$$

Writing out all eight line equations:

$$\begin{aligned} r_1 + r_2 + r_3 &= 0, \\ r_4 + r_5 &= 0, \\ r_6 + r_7 + r_8 &= 0, \\ r_1 + r_4 + r_6 &= 0, \\ r_2 + r_7 &= 0, \\ r_3 + r_5 + r_8 &= 0, \\ r_1 + r_8 &= 0, \\ r_3 + r_6 &= 0. \end{aligned}$$

These eight equations form a homogeneous linear system These eight equations form a homogeneous linear system

$$A\mathbf{r} = 0, \qquad \mathbf{r} = (r_1, \dots, r_8)^T.$$

A direct row reduction shows:

$$\operatorname{rank}(A) = 4, \qquad \dim(\ker A) = 8 - 4 = 4.$$

Thus the entire magic square (with center fixed) is determined by four free parameters. A convenient choice is

$$r_1, \quad r_2, \quad r_3, \quad r_6.$$

Solving the system expresses the remaining four variables as:

This shows that the space of all real 3×3 magic squares with center fixed is a 4-dimensional affine subspace in the $(r_1, \ldots, r_8)$-space. However, a general 3×3 magic square of real numbers (up to affine transformations) has only *two* degrees of freedom. Requiring that each entry $C^2 + r_i$ be a *distinct perfect square* forces the intersection of:

$$\text{a 4-dimensional linear space} \quad \cap \quad \text{a 2-dimensional affine family} \quad \cap \quad \text{a discrete set of integer square points}$$

to be empty. This dimension mismatch is a fundamental obstruction: the linear constraints of the magic square and the arithmetic constraints of being perfect squares cannot be simultaneously satisfied. Therefore, a 3×3 magic square of distinct perfect squares cannot exist.